

DS	Session	Graph	Question Information	Level 1	Challenge 88
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Question description

You are given a tree consisting of n nodes. A *matching* is a set of edges where each node is an endpoint of at most one edge. What is the maximum number of edges in a matching?

Input Format

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n-1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output Format

Print one integer: the maximum number of pairs.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

2

Explanation: One possible matching is (1,2) and (3,4)

Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2
1 3
3 4
3 5

EXPECTED OUTPUT

2

Test Case 2

INPUT (STDIN)

5
1 1
1 2
2 3
3 4

EXPECTED OUTPUT

6

Mandatory Test Cases

Test Case 1

KEYWORD

void dfs(int p,int i)

Test Case 2

KEYWORD

void link(int i,int j)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

350

Test Case 3

NLOC

50

Graph8.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAXN 200005
5
6 int head[MAXN], to[MAXN * 2], nxt[MAXN * 2], edge_cnt;
7 int matched[MAXN];
8 int ans = 0;
9
10 void link(int i, int j) {
11     to[++edge_cnt] = j;
12     nxt[edge_cnt] = head[i];
13     head[i] = edge_cnt;
14 }
15
16 void dfs(int p, int i) {
17     for (int e = head[i]; e; e = nxt[e]) {
18         int v = to[e];
19         if (v != p) {
20             dfs(i, v);
21         }
22     }
23     if (p != 0 && !matched[i] && !matched[p]) {
24         matched[i] = 1;
25         matched[p] = 1;
```

main | CH.SC.U4CSE25225 | Graph8.c

Ln 43, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     ans++;
27 }
28}
29
30int main() {
31     int n;
32     if (scanf("%d", &n) != 1) return 0;
33     for (int i = 0; i < n - 1; i++) {
34         int u, v;
```

```
35     scanf("%d %d", &u, &v);
36     link(u, v);
37     link(v, u);
38 }
39 dfs(0, 1);
40 printf("%d\n", ans);
41 return 0;
42}
43
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5

1 2

1 3

3 4

3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Graph8.exe

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

5

1 1

1 2

2 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Graph8.exe

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>